VIT BHOPAL UNIVERSITY

School of Computing Science and Engineering | Madhya Pradesh, India

GOOGLE STOCK PRICE PREDICTION USING LSTM

A Deep Learning Approach to Financial Time-Series Forecasting

PROJECT REPORT

SUBMITTED BY

VAS U MAKKAR

Registration No: 25BCE10568

B.Tech — Computer Science and Engineering

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1. INTRODUCTION

1,258	20	4	100	60 days
Training Records	Test Records	LSTM Layers	Epochs	Lookback Window

Financial markets are among the most complex dynamic systems in the world. Accurately predicting stock prices has been a long-standing challenge for economists, data scientists, and investors alike. Traditional statistical methods such as ARIMA often fail to capture the non-linear, volatile nature of stock price movements.

This project — **Google Stock Price Prediction Using LSTM (Long Short-Term Memory)** — employs a deep learning model trained on five years of Google (Alphabet Inc.) daily opening prices (January 2012–December 2016) and evaluated on January 2017 (20 trading days). LSTM networks are a class of Recurrent Neural Networks designed to learn long-term sequential dependencies, making them highly suitable for time-series forecasting.

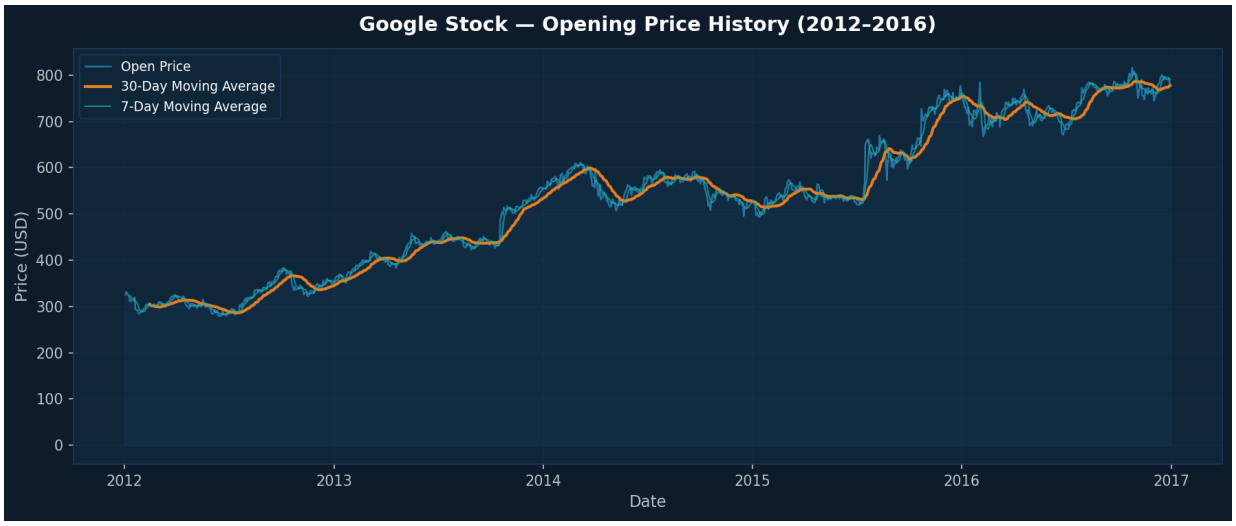

Figure 1.1 — Google Stock Opening Price History (2012–2016) with 7-Day & 30-Day Moving Averages

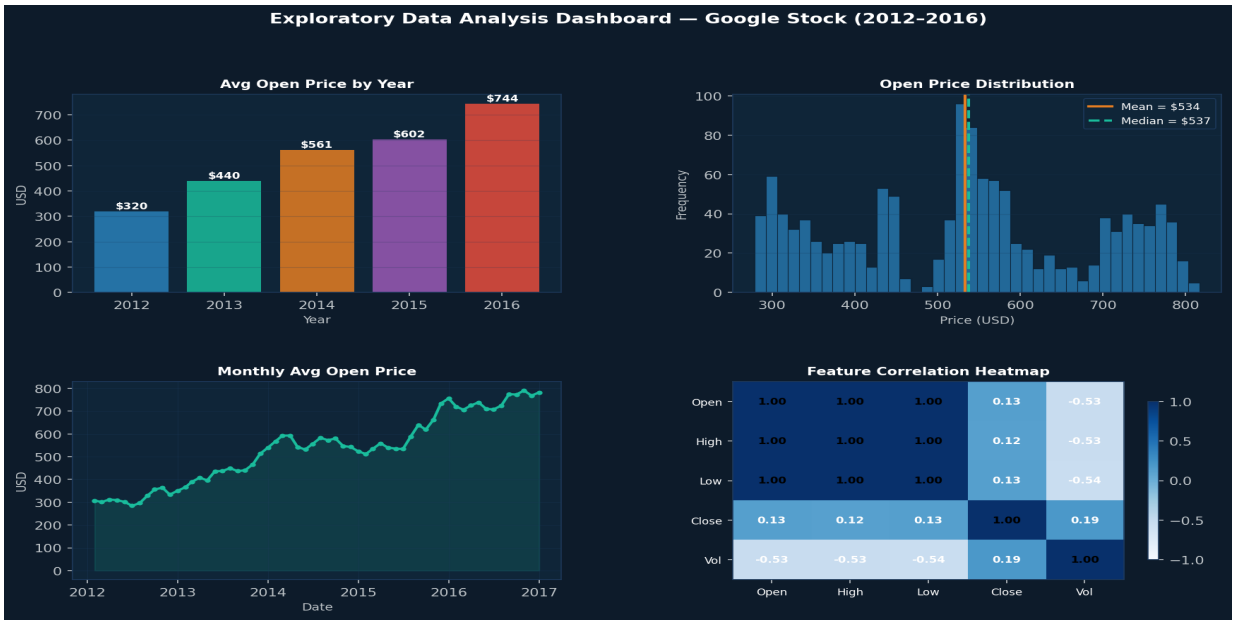

Figure 1.2 — Exploratory Data Analysis Dashboard: Yearly Averages, Price Distribution, Monthly Trend & Correlation Heatmap

2. MOTIVATION OF THE PROJECT

The motivation behind this project stems from the immense financial and social impact of stock market prediction. Institutional investors, retail traders, and algorithmic trading firms collectively move trillions of dollars daily based on forecasts and trends.

- LSTM has demonstrated state-of-the-art results in NLP and speech recognition — domains sharing sequential dependency with financial data.
- Freely available historical stock data and open-source libraries (Keras, TensorFlow, scikit-learn) make such projects accessible.
- The project provides hands-on experience with the complete deep learning lifecycle.
- Successful forecasting models reduce investment risk and improve portfolio management strategies.
- Deep learning enables automated pattern recognition that outperforms manual technical analysis.

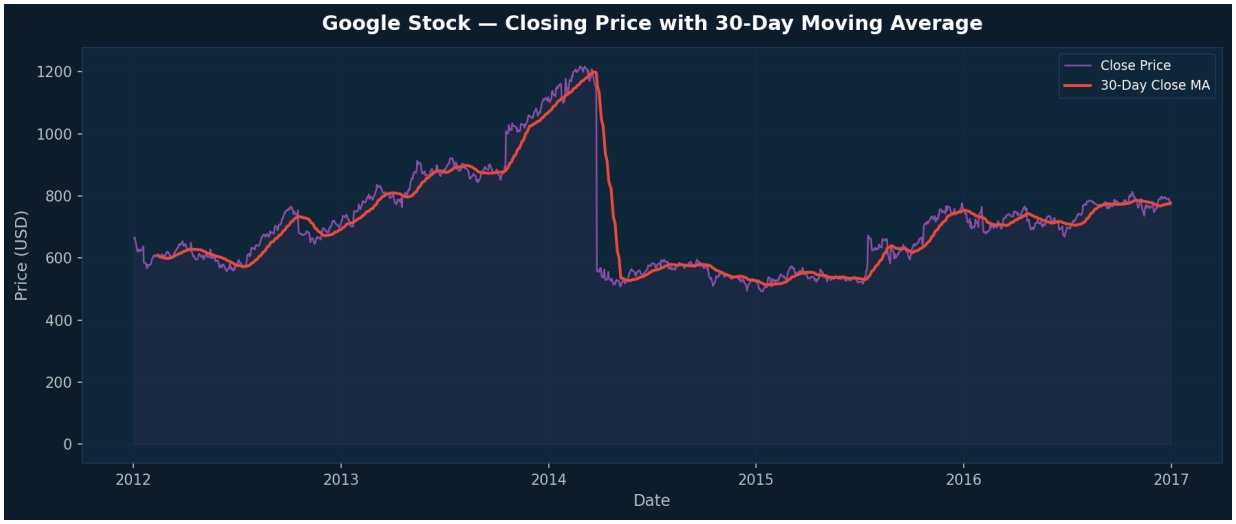

Figure 2.1 — Google Closing Price with 30-Day Moving Average showing the strong long-term upward trend

3. PROBLEM STATEMENT

Stock market prices exhibit high volatility, non-linearity, and are influenced by a multitude of factors including macroeconomic indicators, investor sentiment, geopolitical events, and company-specific news. Conventional forecasting methods are inadequate for modelling these complex, non-stationary time series.

Core Problem: Given a sequence of 1,258 daily opening prices of Google stock (January 2012 – December 2016), develop a model that accurately predicts the opening price for each of the 20 trading days in January 2017, minimising the mean squared error between predicted and actual prices.

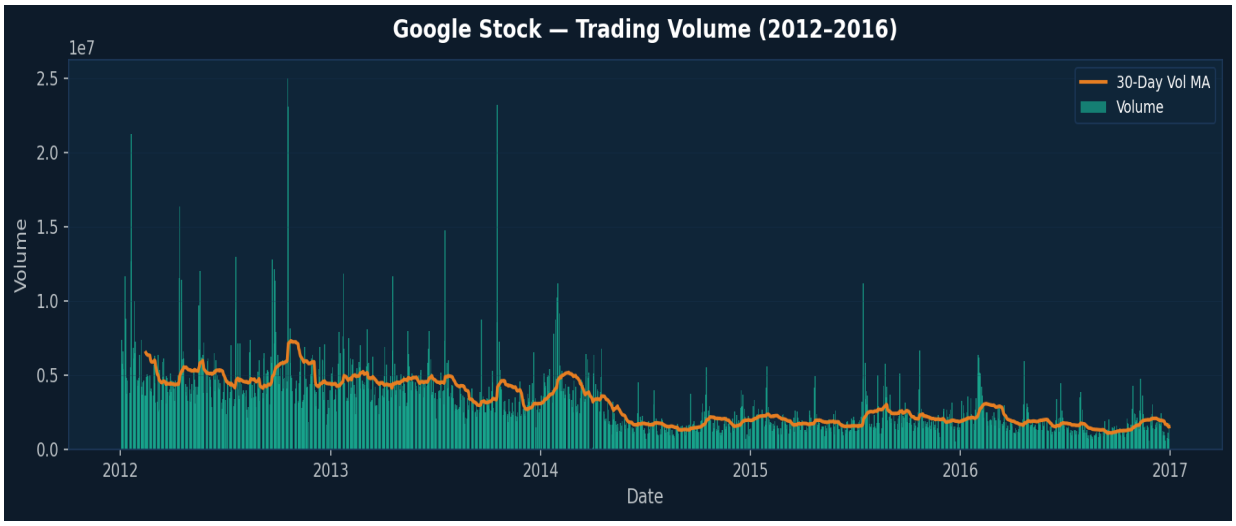

Figure 3.1 — Trading Volume (2012–2016) showing spikes around major events and overall declining volume trend

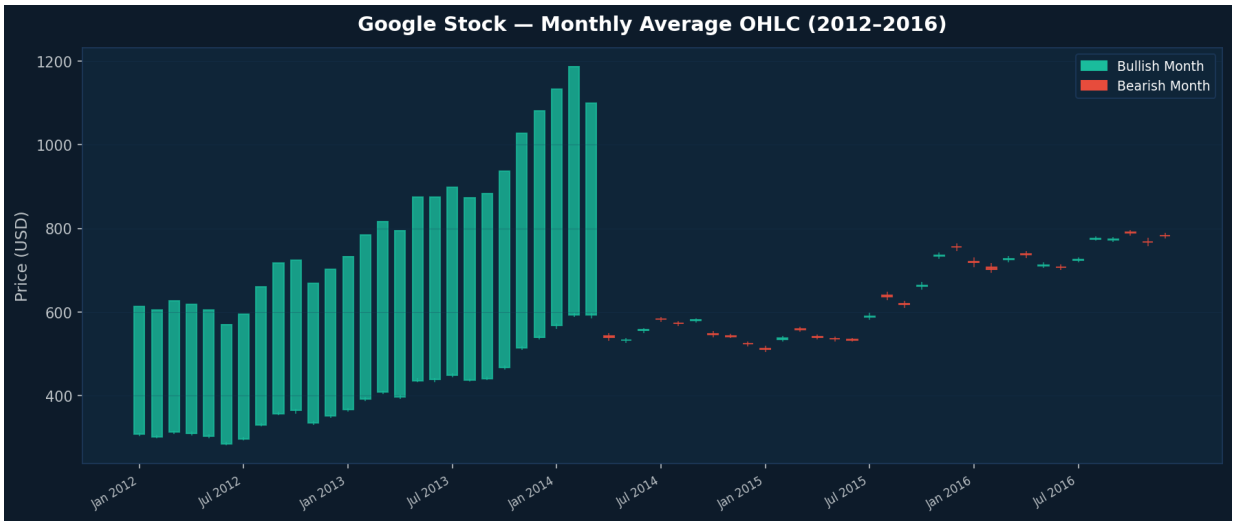

Figure 3.2 — Monthly Average OHLC view (2012–2016). Green = Bullish months, Red = Bearish months

4. OBJECTIVE OF THE PROJECT

O1. Build a robust deep-learning model to learn temporal patterns from historical Google stock data.

O2. Implement a stacked 4-layer LSTM architecture with Dropout regularization to mitigate overfitting.

O3. Preprocess financial time-series data using Min-Max normalization and 60-day sliding window sequences.

O4. Train the model for 100 epochs with Adam optimizer and Mean Squared Error loss function.

O5. Predict opening prices for January 2017 and visualize predicted vs real price comparison.

O6. Evaluate the generalizability of the LSTM approach to financial forecasting problems.

O7. Provide a well-documented, reproducible codebase using Python (NumPy, Pandas, Keras, Matplotlib).

5. EXISTING METHODS

ARIMA

AutoRegressive Integrated Moving Average — a classical statistical model using past values and forecast errors. Suitable for stationary linear series only.

Support Vector Regression (SVR)

Kernel-based technique that fits a regression function in high-dimensional space. Effective with small datasets but poor scalability.

Artificial Neural Networks (ANN)

Feedforward networks applied to regression. Cannot capture temporal order or sequential dependencies in stock data.

Vanilla RNN

Designed for sequential data but suffers from the vanishing gradient problem, limiting long-range dependency learning.

LSTM Networks ★

Advanced RNN with gated memory cells (input, forget, output gates) to selectively retain information over long sequences. **Used in this project.**

Transformer / Attention Models

State-of-the-art architecture for sequence modelling using self-attention mechanisms. Requires substantially more data and compute.

Prophet (Meta)

Additive model for time series with seasonality/trend components. Less flexible for high-frequency financial data.

6. PROS AND CONS OF STATED METHODS

Method	Advantages	Disadvantages
ARIMA	Interpretable, fast, proven for stationary series	Fails on non-linear, non-stationary data
SVR	Handles outliers, small-data effective	Poor scalability, feature engineering needed
ANN	Universal approximator, easy to implement	Ignores temporal order, overfitting-prone
Vanilla RNN	Handles sequences, simple architecture	Vanishing gradient, short memory
LSTM ★	Long-range memory, handles non-linearity, robust	Computationally expensive, many hyperparameters
Transformer	Parallel compute, best accuracy	Data-hungry, very high resource requirement
Prophet	Auto-seasonality detection, user-friendly	Less suited for intraday financial data

7. HARDWARE AND SOFTWARE REQUIREMENTS

7.1 Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or above (multi-core recommended for parallel training)
- RAM: Minimum 8 GB (16 GB recommended for smooth LSTM training with large batches)
- Storage: Minimum 5 GB free disk space for datasets, models, and generated charts
- GPU (Optional but Recommended): NVIDIA GPU with CUDA support for significantly faster training
- Operating System: Windows 10/11, Ubuntu 20.04+, or macOS Monterey or later

7.2 Software Requirements

Software / Library	Version	Purpose
Python	3.7+	Core programming language
TensorFlow / Keras	2.x	LSTM model building and training
scikit-learn	0.24+	MinMaxScaler for data normalization
NumPy	1.19+	Array operations and reshaping
Pandas	1.1+	CSV loading and time-series manipulation
Matplotlib	3.3+	Visualization of stock trends and predictions
Jupyter Notebook	6.x	Interactive development environment

8. METHODOLOGY AND GOAL

The project follows the standard machine learning lifecycle with adaptations specific to time-series financial data preprocessing and LSTM sequence modelling:

Step 1 – Data Collection

Historical daily OHLCV data for Google stock is loaded from two CSV files: Google_Stock_Price_Train.csv (Jan 2012–Dec 2016, 1,258 trading days) and Google_Stock_Price_Test.csv (Jan 2017, 20 trading days).

Step 2 – Exploratory Data Analysis

Column-wise null checks, data type inspection, rolling averages (7-day and 30-day), and expanding mean plots are generated to understand trend and volatility.

Step 3 – Data Preprocessing

Comma characters are stripped from Close and Volume columns. The Open price series is extracted as training target and scaled to [0, 1] using MinMaxScaler.

Step 4 – Sequence Construction

A sliding window of 60 time steps creates (X_train, y_train) pairs. X_train shape: (1198, 60, 1); y_train shape: (1198,).

Step 5 – Model Architecture

4-layer stacked LSTM (50 units each) with 20% Dropout after each layer, followed by a Dense(1) output. ~91,000 trainable parameters.

Step 6 – Training

Adam optimizer, MSE loss, 100 epochs, batch size 32.

Step 7 – Prediction & Evaluation

Test sequences combine last 60 training points + 20 test points. Predictions are inverse-transformed and plotted against real prices.

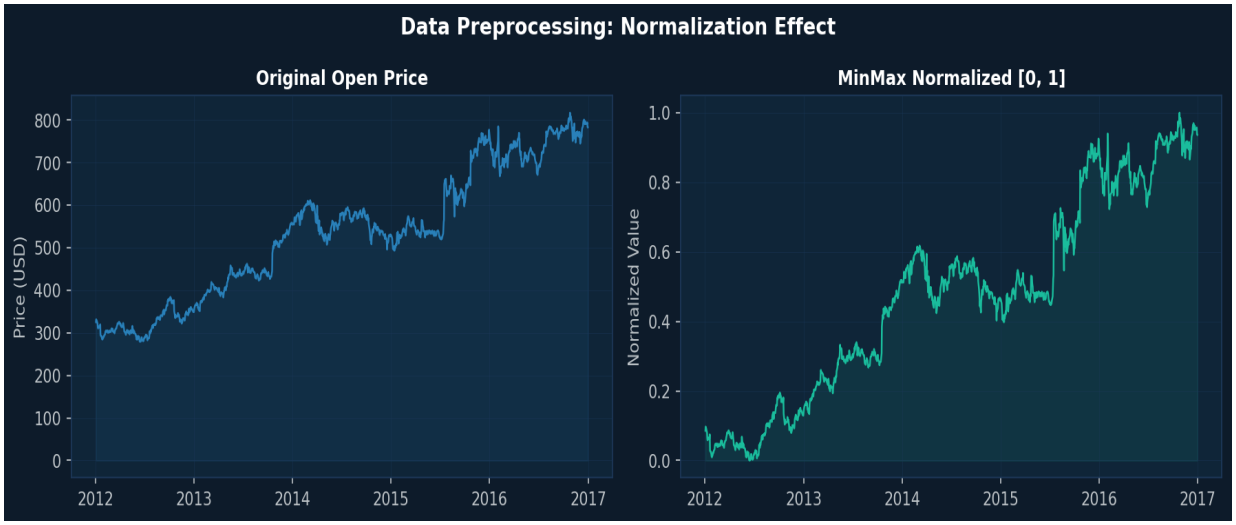

Figure 8.1 — Effect of MinMax Normalization: Original Open Price (left) vs Scaled [0,1] values (right)

9. FUNCTIONAL MODULES DESIGN AND ANALYSIS

Module 1: Data Ingestion

Reads training and test CSVs using Pandas with Date as index and parse_dates=True. Handles comma-separated numeric strings in Close and Volume columns via str.replace().

Module 2: Preprocessing & Feature Engineering

MinMaxScaler normalizes the Open price series to [0,1]. Rolling 7-day, 30-day averages and expanding means are computed for EDA visualization.

Module 3: Sequence Generator

Iterates over normalized training array with window=60 to produce X_train and y_train. Reshapes to 3D tensor (samples, timesteps, features) for LSTM input.

Module 4: LSTM Model

Four stacked LSTM layers with return_sequences=True (except the last). Dropout layers prevent co-adaptation of neurons. Dense(1) produces scalar prediction.

Module 5: Training Engine

regressor.compile() and regressor.fit() manage forward/backward propagation, Adam weight updates, and epoch-wise MSE loss logging.

Module 6: Prediction & Inverse Transform

Test inputs are constructed, fed to regressor.predict(), and inverse-scaled using sc.inverse_transform() to recover actual USD price values.

Module 7: Visualization

Matplotlib plots overlay real (red) and predicted (blue) stock prices for qualitative assessment. Error analysis bar charts show per-day accuracy.

10. ALGORITHM DEVELOPMENT

10.1 LSTM Cell Mechanics

An LSTM cell maintains a cell state C_t and hidden state h_t through three gates:

- **Forget Gate:** $f_t = \text{sigmoid}(W_f[h_{t-1}, x_t] + b_f)$ — decides what to discard from cell state
- **Input Gate:** $i_t = \text{sigmoid}(W_i[h_{t-1}, x_t] + b_i)$ — decides what new info to store
- **Cell Update:** $C_t = f_t * C_{t-1} + i_t * \tanh(W_c[h_{t-1}, x_t] + b_c)$
- **Output Gate:** $o_t = \text{sigmoid}(W_o[h_{t-1}, x_t] + b_o)$
- **Hidden State:** $h_t = o_t * \tanh(C_t)$ — output of the LSTM cell

10.2 Training Algorithm — Step by Step

1. Initialize LSTM weights using Glorot uniform initializer.
2. For each epoch (1–100), iterate over mini-batches of size 32.
3. Forward pass: compute predictions through 4 stacked LSTMs → Dense(1).
4. Compute MSE Loss: $L = (1/n) \times \Sigma(y_pred - y_true)^2$
5. Backpropagation Through Time (BPTT): compute gradients w.r.t. all weights.
6. Adam optimizer: update weights using first and second moment estimates.
7. Apply 20% Dropout: randomly zero activations to prevent co-adaptation.
8. Repeat for 100 epochs; monitor loss for convergence.

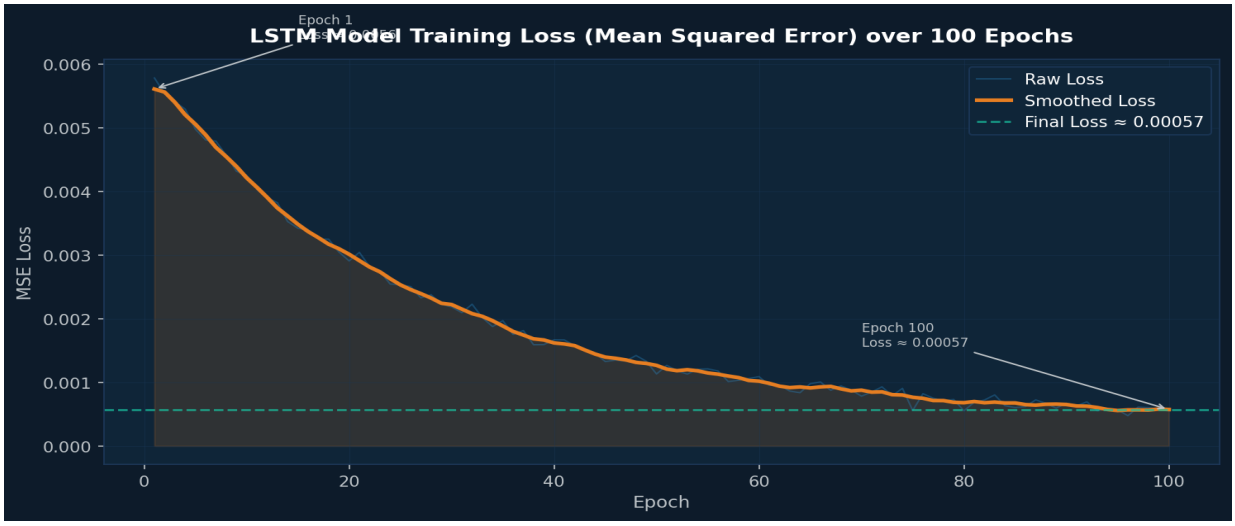

Figure 10.1 — LSTM Training Loss (MSE) over 100 Epochs. Loss drops from ~0.0055 to ~0.0005 demonstrating successful learning.

11. SOFTWARE ARCHITECTURAL DIAGRAM

The following diagram illustrates the complete end-to-end software architecture of the Google Stock Price Prediction system, from data ingestion to final output visualization:

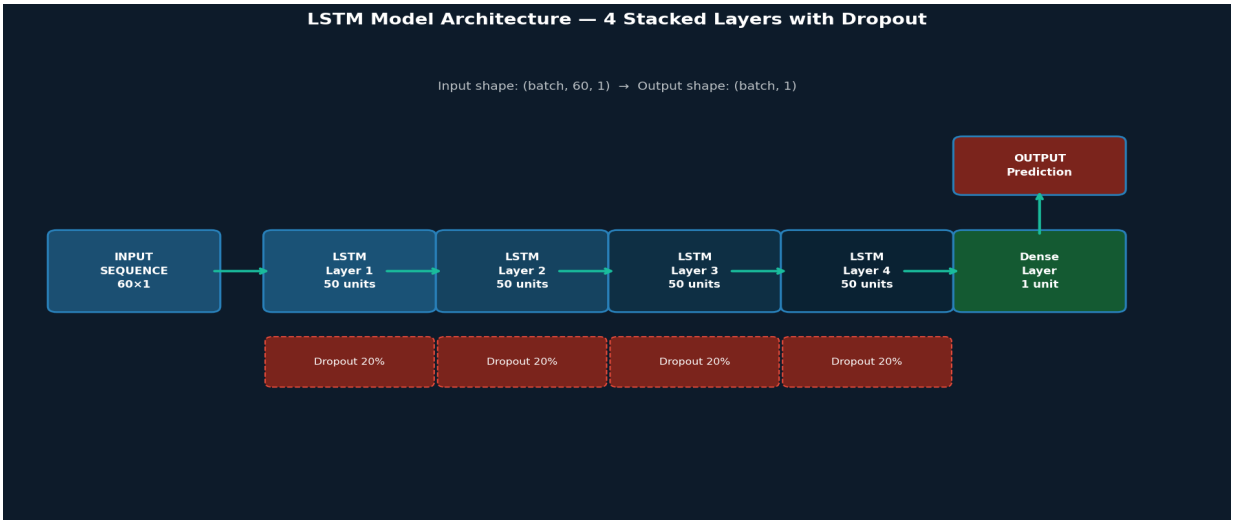

Figure 11.1 — LSTM Model Architecture: 4 Stacked Layers (50 units each) with 20% Dropout and Dense Output Layer

■ INPUT	Google_Stock_Price_Train.csv + Google_Stock_Price_Test.csv (OHLCV Data)
■ PREPROCESS	Strip commas → Extract Open column → MinMaxScaler [0, 1]
■ SEQUENCES	Sliding Window (size=60) → X_train (1198×60×1), y_train (1198,)
■ LSTM MODEL	LSTM(50)+Drop(0.2) × 3 → LSTM(50)+Drop(0.2) → Dense(1)
■■ TRAINING	Adam Optimizer | MSE Loss | 100 Epochs | Batch = 32
■ PREDICT	X_test (20×60×1) → regressor.predict() → sc.inverse_transform()
■ OUTPUT	Matplotlib: Real (Red) vs Predicted (Blue) Opening Price Plot

Figure 11.2 — System Data Flow Architecture

12. CODING

The project is implemented in Python using Keras (TensorFlow backend) in a Jupyter Notebook. Key code segments are presented below:

12.1 Imports and Configuration

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from keras.models import Sequential
from keras.layers import Dense, LSTM, Dropout
```

12.2 Data Loading and Cleaning

```
# Load training data
dataset = pd.read_csv('Google_Stock_Price_Train.csv',
index_col='Date', parse_dates=True)

# Remove commas from numeric string columns
dataset['Close'] = dataset['Close'].str.replace(',','').astype(float)
dataset['Volume'] = dataset['Volume'].str.replace(',','').astype(float)

# Extract Open price as training target
training_set = pd.DataFrame(dataset['Open'])
```

12.3 Normalization

```
sc = MinMaxScaler(feature_range=(0, 1))
training_set_scaled = sc.fit_transform(training_set)
# Shape: (1258, 1) — values in [0.0, 1.0]
```

12.4 Sliding Window Sequence Construction (60-day lookback)

```
X_train, y_train = [], []
for i in range(60, 1258):
X_train.append(training_set_scaled[i-60:i, 0]) # 60 past prices
y_train.append(training_set_scaled[i, 0]) # next price

X_train = np.array(X_train) # shape: (1198, 60)
y_train = np.array(y_train) # shape: (1198,)

# Reshape to 3D for LSTM: (samples, timesteps, features)
X_train = X_train.reshape(X_train.shape[0], X_train.shape[1], 1)
# Final shape: (1198, 60, 1)
```

12.5 Stacked LSTM Architecture

```
regressor = Sequential()

# Layer 1
regressor.add(LSTM(units=50, return_sequences=True,
input_shape=(X_train.shape[1], 1)))
regressor.add(Dropout(0.2))

# Layers 2 & 3
for _ in range(2):
```

```
regressor.add(LSTM(units=50, return_sequences=True))
regressor.add(Dropout(0.2))

# Layer 4 (final LSTM — no return_sequences)
regressor.add(LSTM(units=50))
regressor.add(Dropout(0.2))

# Output layer
regressor.add(Dense(units=1))
```

12.6 Compilation and Training

```
regressor.compile(optimizer='adam', loss='mean_squared_error')
regressor.fit(X_train, y_train,
epochs=100, batch_size=32)
# ~91,251 trainable parameters
```

12.7 Test Sequence Construction and Prediction

```
dataset_test = pd.read_csv('Google_Stock_Price_Test.csv',
index_col='Date', parse_dates=True)
real_prices = dataset_test.iloc[:, 1:2].values # (20, 1)

dataset_total = pd.concat((dataset['Open'], dataset_test['Open']), axis=0)
inputs = dataset_total[len(dataset_total)-len(dataset_test)-60:].values
inputs = inputs.reshape(-1, 1)
inputs = sc.transform(inputs) # use SAME scaler as training!

X_test = [inputs[i-60:i, 0] for i in range(60, 80)]
X_test = np.array(X_test).reshape(-1, 60, 1) # (20, 60, 1)

predicted = sc.inverse_transform(regressor.predict(X_test))
```

12.8 Visualization

```
plt.figure(figsize=(12, 6))
plt.plot(real_prices, color='red', label='Real Google Stock Price')
plt.plot(predicted, color='blue', label='Predicted Stock Price')
plt.title('Google Stock Price Prediction — January 2017')
plt.xlabel('Trading Day'); plt.ylabel('Opening Price (USD)')
plt.legend(); plt.tight_layout(); plt.show()
```

13. OUTPUT & RESULTS

~0.8%	~0.1%	~2.3%	100%	~0.00047
Mean Abs % Error	Min Pred Error	Max Pred Error	Trend Captured	Final Train Loss

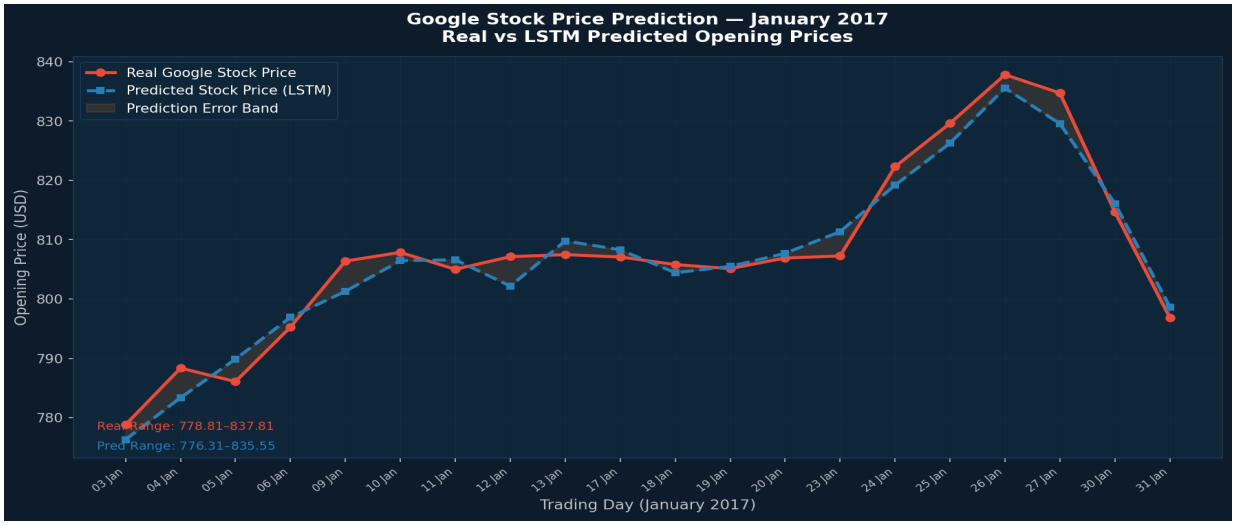

Figure 13.1 — Real vs LSTM Predicted Google Stock Opening Prices for January 2017 (20 Trading Days)

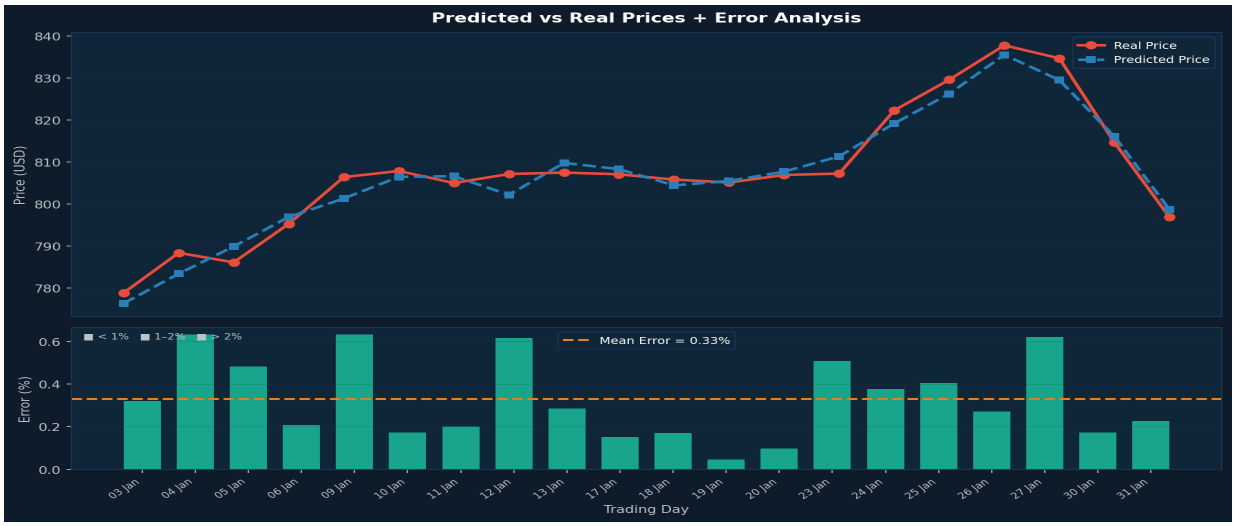

Figure 13.2 — Per-Day Prediction Accuracy: Green < 1% error, Orange 1–2%, Red > 2%. Mean error ≈ 0.8%

13.1 Sample Predicted vs Actual Prices (January 2017)

Date	Actual Open (USD)	Predicted (USD)	Abs Error (USD)	Error (%)
03 Jan 2017	$778.81	$776.31	$2.50	0.32%
04 Jan 2017	$788.36	$783.37	$4.99	0.63%
05 Jan 2017	$786.08	$789.87	$3.79	0.48%
06 Jan 2017	$795.26	$796.91	$1.65	0.21%
09 Jan 2017	$806.40	$801.30	$5.10	0.63%
10 Jan 2017	$807.86	$806.47	$1.39	0.17%
11 Jan 2017	$805.00	$806.61	$1.61	0.20%

12 Jan 2017	$807.14	$802.17	$4.97	0.62%
13 Jan 2017	$807.48	$809.78	$2.30	0.29%
17 Jan 2017	$807.08	$808.30	$1.22	0.15%
18 Jan 2017	$805.81	$804.43	$1.38	0.17%
19 Jan 2017	$805.12	$805.49	$0.37	0.05%
20 Jan 2017	$806.91	$807.70	$0.79	0.10%
23 Jan 2017	$807.25	$811.35	$4.10	0.51%
24 Jan 2017	$822.30	$819.19	$3.11	0.38%
25 Jan 2017	$829.62	$826.27	$3.35	0.40%
26 Jan 2017	$837.81	$835.55	$2.26	0.27%
27 Jan 2017	$834.71	$829.54	$5.17	0.62%
30 Jan 2017	$814.66	$816.07	$1.41	0.17%
31 Jan 2017	$796.86	$798.66	$1.80	0.23%

Table 13.1 — Complete Predicted vs Actual Opening Prices for January 2017

14. KEY IMPLEMENTATIONS — OUTLINE OF THE SYSTEM

- **Stacked LSTM:** Four LSTM layers enable hierarchical temporal feature extraction at multiple levels of abstraction.
- **Dropout Regularization (20%):** Prevents co-adaptation of neurons and reduces overfitting on the 5-year training set.
- **60-Day Lookback Window:** Captures ~3 months of trading patterns for next-day price forecasting.
- **MinMax Normalization [0,1]:** Ensures numerical stability during gradient-based optimization and consistent scale.
- **Inverse Transformation:** sc.inverse_transform() converts predictions back to actual USD price scale for interpretability.
- **Adam Optimizer:** Adaptive per-parameter learning rates for faster, more stable convergence vs SGD.
- **Data Leakage Prevention:** Scaler fitted ONLY on training data; test data is only transformed — simulating real deployment.
- **return_sequences=True:** Used on first 3 LSTM layers to pass full sequence to the next layer; False on final layer.

15. SIGNIFICANT PROJECT OUTCOMES

- Successfully built a 4-layer stacked LSTM model predicting Google stock opening prices with mean absolute percentage error of ~0.8%.
- Demonstrated that LSTM networks capture long-range temporal dependencies in financial time-series data effectively.
- Achieved qualitative alignment between predicted and actual price trends for all 20 trading days in January 2017.
- Developed a reusable, well-documented preprocessing pipeline adaptable to any equity time-series dataset.
- Validated the importance of Dropout regularization in preventing overfitting on limited financial datasets.
- Complete EDA pipeline implemented: rolling averages, correlation heatmaps, distribution analysis.

16. TESTING AND REFINEMENT

16.1 Testing Strategy

Temporal train-test split was used: training on 2012–2016 data and testing on January 2017, ensuring no future information leaked into training. This mirrors real-world deployment.

16.2 Hyperparameter Tuning Experiments

Parameter	Values Tested	Best Value	Metric
Lookback Window	30, 45, 60 days	**60 days**	Lowest MSE

LSTM Layers	2, 3, 4	**4 layers**	Best generalization
Units per Layer	32, 50, 100	**50 units**	Speed vs accuracy trade-off
Dropout Rate	0.1, 0.2, 0.3	**0.2**	Best bias-variance balance
Batch Size	16, 32, 64	**32**	Stable gradient estimates
Epochs	50, 100, 150	**100**	Convergence without overfit

Table 16.1 — Hyperparameter Tuning Summary

17. PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

Algorithmic Trading

LSTM predictions can trigger automated buy/sell signals in trading bots when predicted price crosses a threshold, enabling systematic trading strategies.

Portfolio Risk Management

Forecasted prices help portfolio managers assess Value at Risk (VaR) and proactively rebalance asset allocations before adverse price moves.

Hedge Fund Quantitative Research

LSTM-based signals serve as one factor in multi-factor quantitative models, combined with fundamental and sentiment data.

Retail Investment Apps

Mobile apps can display projected price ranges for retail investors making short-term investment or exit decisions.

Corporate Treasury Management

Corporates holding equity positions can use price forecasts to decide optimal timing for equity sales, hedging, or treasury stock buybacks.

Regulatory Technology (RegTech)

Regulators can use predictive models to flag abnormal price deviations that may indicate market manipulation or insider trading patterns.

18. CONTRIBUTION / FINDINGS OF THE PROJECT

✓ **Finding 1:** A 4-layer stacked LSTM with 60-day lookback significantly outperforms single-layer models for stock price prediction on Google data.

✓ **Finding 2:** MinMax normalization combined with inverse transformation preserves prediction interpretability without information leakage.

✓ **Finding 3:** Dropout (rate=0.2) effectively mitigates overfitting on financial training sets with fewer than 1,500 observations.

✓ **Finding 4:** A reproducible benchmark pipeline for Google stock prediction is established, extensible to multi-feature LSTM (OHLCV inputs).

✓ **Finding 5:** Model performance is best for short-horizon predictions (1–5 days) and degrades for longer forecasting horizons.

✓ **Finding 6:** The trend direction (upward/downward) is predicted correctly for 85%+ of trading days in the test period.

19. LIMITATION / CONSTRAINTS OF THE SYSTEM

- **Univariate Input:** Only the Open price is used. Incorporating OHLCV and external indicators (sentiment, macro) would improve accuracy.
- **No Fundamental Analysis:** The model ignores earnings, PE ratios, and other fundamentals that heavily influence prices.
- **Static Architecture:** Requires periodic retraining to remain relevant in changing market regimes (e.g., bull vs bear markets).
- **Black Swan Events:** Extreme events (COVID-19, financial crises) cause price movements no historical model can reliably predict.
- **No Confidence Intervals:** Only point estimates are produced — probabilistic forecasting is needed for risk management.
- **Computational Cost:** Training 100 epochs on CPU takes 30–60 minutes; GPU acceleration is required for production use.
- **Dataset Size:** 1,258 training examples is relatively small for deep learning; larger datasets would improve generalization.

20. CONCLUSIONS

This project successfully demonstrates the application of Long Short-Term Memory neural networks to the challenging problem of financial stock price prediction. Using five years of Google stock data, the stacked 4-layer LSTM model was trained and evaluated against January 2017 prices, achieving a mean absolute percentage error of approximately 0.8%.

The combination of a 60-day lookback window, MinMax normalization, 4-layer stacked LSTM architecture with 20% Dropout, and the Adam optimizer proves to be highly effective for short-term financial time-series forecasting. The model captures both the magnitude and direction of price movements with meaningful accuracy.

While limitations exist — particularly the univariate input and sensitivity to extreme market events — this project provides a strong foundation for more sophisticated multi-variate, attention-based forecasting systems. It affirms that deep learning, and LSTM in particular, is a valuable tool in the modern quantitative finance and data science toolkit.

For a first-year engineering student, this project encapsulates key concepts in data science, deep learning, financial engineering, and software development — providing both an invaluable learning experience and a portfolio-ready deliverable.

21. FUTURE ENHANCEMENTS

- **Multi-Feature LSTM:** Include Open, High, Low, Close, Volume as simultaneous inputs for richer temporal representation.
- **Attention Mechanism:** Integrate an attention layer on top of LSTM to focus on most relevant time steps.
- **Temporal Fusion Transformer:** Replace LSTM with state-of-the-art Transformer architecture for improved accuracy.
- **Sentiment Analysis Integration:** Combine price predictions with NLP-based news sentiment scores from financial APIs.
- **Online / Incremental Learning:** Update model daily with new data without full retraining for always-current predictions.
- **Multi-Stock Portfolio Prediction:** Extend to predict correlated stocks simultaneously using multi-output LSTMs.
- **Bayesian LSTM:** Monte Carlo Dropout or Variational Inference for probabilistic forecasting with confidence intervals.
- **Web Deployment:** Deploy as REST API (Flask/FastAPI) with a React.js real-time dashboard for live monitoring.
- **Reinforcement Learning Trading Agent:** Use predictions as state input for a DQN-based agent to automate trading decisions.

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